

AS/NZS 4936:2002

Reconfirmed 2020

Australian/New Zealand Standard™

**Air admittance valves (AAVs) for use in  
sanitary plumbing and drainage  
systems**

## AS/NZS 4936:2002

This Joint Australian/New Zealand Standard was prepared by Joint Technical Committee WS-002, Sanitary Plumbing Fittings. It was approved on behalf of the Council of Standards Australia on 14 August, 2002 and on behalf of the Council of Standards New Zealand on 24 July 2002. It was published on 2 September 2002.

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STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

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**RECONFIRMATION**  
**OF**  
**AS/NZS 4936:2002**  
**Air admittance valves (AAVs) for use in**  
**sanitary plumbing and drainage systems**

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Major stakeholders of this publication have reviewed the content of this publication and in accordance with Standards Australia procedures for reconfirmation, it has been determined that the publication is still valid and does not require change.

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NOTES

# Australian/New Zealand Standard™

## **Air admittance valves (AAVs) for use in sanitary plumbing and drainage systems**

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## PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee WS-002, Sanitary Plumbing Fittings.

The objective of this Standard is to provide performance requirements for air admittance valves suitable for use in plumbing systems designed in accordance with AS/NZS 3500.2.1 and AS/NZS 3500.2.2.

This Standard takes cognisance of prEN 12380.2-2001, *Air admittance valves — For use in drainage systems*, from which it has been generally adapted but extends its scope to include those air admittance valves that are integral with the fixture trap.

The term ‘normative’ has been used in this Standard to define the application of the appendix to which it applies. A ‘normative’ appendix is an integral part of a Standard.

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## STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

**Australian/New Zealand Standard****Air admittance valves (AAVs) for use in sanitary plumbing and drainage systems****1 SCOPE**

This Standard specifies the requirements and test methods for air admittance valves including those that are integral to a fixture trap. This Standard does not cover air admittance valves for use where the air temperature is either below 0°C or above 60°C.

**2 REFERENCED DOCUMENTS**

The following documents are referred to in this Standard:

**AS**

|          |                                                                   |
|----------|-------------------------------------------------------------------|
| 2888     | Methods of testing plastics waste fittings                        |
| 2888.8   | Thermal cycling test                                              |
| 3500     | National plumbing and drainage                                    |
| 3500.2.1 | Part 2.1: Sanitary plumbing and drainage—Performance Requirements |

**AS/NZS**

|          |                                                               |
|----------|---------------------------------------------------------------|
| 3500     | National plumbing and drainage                                |
| 3500.0   | Part 0: Glossary of terms                                     |
| 3500.2.2 | Part 2.2: Sanitary plumbing and drainage—Acceptable solutions |

**EN**

|         |                                                   |
|---------|---------------------------------------------------|
| 12380.2 | Air admittance valves—For use in drainage systems |
|---------|---------------------------------------------------|

**3 DEFINITIONS**

For the purpose of this Standard the definitions below apply.

**3.1 Air admittance valve**

A component fitted to a plumbing system allowing air to enter the piping system, but not allowing air or gases out.

**3.2 Determined airflow capacity**

The airflow, in litres per second, measured on the apparatus described in Appendix A of this Standard whilst producing a pressure of  $-250 \pm 10$  Pa at the manometer tapping.

**4 MARKING**

Each valve shall be marked with the following information using means that do not interfere with the valve's function:

- The manufacturer's name, brand or trademark.
- The number of this Standard.
- The mark of the certifying body.
- The determined airflow capacity of the valve in litres per second.



- (e) The maximum fixture unit loadings of—
  - (i) discharge pipes and
  - (ii) discharge stacks that the valve may use to vent.
- (f) Suitability for use in direct sunlight.

Certification of resistance to ultra-violet by the supplier of the material used for manufacture of the valve body shall be sufficient for verification of this claim

## 5 CLASSIFICATION

Air admittance valves complying with this Standard are suitable for location below the flood level of the appliance or fixture and suitable for operation in the temperature range 0°C to 60°C.

## 6 DESIGN

The connections of air admittance valves shall be dimensionally compatible with the product to which the connection is to be made, as specified in the relevant Standard for the product.

The air admittance valve shall be designed to be installed in accordance with the manufacturer's instructions in respect to jointing methods, the orientation of the valve, minimum clearances from the pipe obvert and provision of access for inspection, maintenance, testing and replacement.

## 7 PERFORMANCE REQUIREMENTS

### 7.1 General

Air admittance valves shall be tested as supplied by the manufacturer, and installed in the manner recommended by the manufacturer. Where air admittance valves are supplied with one or more adaptors, the tests shall be repeated with each adaptor, to ascertain any effect that the adaptor may impose on the performance of the valve. The tests shall be carried out with the air admittance valve mounted within 5° of the vertical and fitted with any insulation, insect screen or other device normally supplied with the valve by the manufacturer of the air admittance valve.

### 7.2 Sampling for tests

Three samples of each type and variation of types shall be selected at random and the same valves subjected to all performance tests.

Sample valves shall be subjected to the tests in the following order:

- (a) Drop test
- (b) Airtightness test
- (c) Endurance and temperature test (on one sample only).
- (d) Opening characteristic and airflow test.
- (e) Thermal cycling test (only for products incorporating both a fixture trap and an air admittance valve).

### 7.3 Durability—Normal handling

Air admittance valves shall be durable under conditions of normal handling. They shall be deemed to comply with this requirement if, when subjected to the drop test in accordance with Appendix B, each valve shows no visible signs of damage or deformation likely to affect the functioning of the valve.

#### **7.4 Airtightness**

Air admittance valves shall withstand a positive internal pressure up to 10 000 Pa without leakage. When tested in accordance with Appendix C, the valves shall show no signs of leakage.

#### **7.5 Endurance and temperature**

Air admittance valves shall operate reliably throughout their service life. They shall be deemed to comply with this requirement if, when tested in accordance with Appendix D, valves pass the airtightness test of Appendix C after operating at the test temperatures over the given time periods.

#### **7.6 Opening characteristic and flow**

Air admittance valves shall have a measurable airflow when a constant pressure of  $-150 \pm 10$  Pa is applied to the inside of the valve. When tested in accordance with Appendix A, the valve shall have an airflow no less than the determined airflow capacity when a constant pressure of  $-250 \pm 10$  Pa is applied to the inside of the valve.

#### **7.7 Thermal cycling test for fixture traps**

Products where the air admittance valve is integral with a fixture trap shall additionally comply with the thermal immersion cycling test of AS 2888.8 for 1 000 cycles at  $20 \pm 2$ ,  $-0^\circ\text{C}$  and  $60 \pm 2$ ,  $-0^\circ\text{C}$  as the upper and lower test temperatures respectively. For this test the product shall be supported or suspended in such a manner that water drains from it between immersions.

### **8 INSTALLATION INSTRUCTIONS**

Each valve shall be packaged with a complete set of installation instructions in accordance with Appendix E.

Valves shall be installed in accordance with AS/NZS 3500.2.2.

## APPENDIX A

### OPENING CHARACTERISTIC AND AIRFLOW TEST

(Normative)

#### A1 SCOPE

This Appendix sets out the method for determining the opening characteristic and airflow in an air admittance valve.

#### A2 PRINCIPLE

The valve is subjected to a suction, to establish the opening characteristics of its mechanism and the airflow is measured at a specified pressure.

#### A3 APPARATUS

A suitable rig (see Figure A1) comprised of—

- (a) an appropriate pipe with a static pressure tapping for a suitable manometer;
- (b) a manometer;
- (c) a variable rated electric fan or equivalent means to induce and maintain an airflow through the pipe; and
- (d) an instrument to measure the volumetric airflow in the pipe.

NOTE: Tests should be conducted in a draught-free constant-temperature environment.

#### A4 PROCEDURE

The procedure shall be as follows:

- (a) Connect the air admittance valve to the top of the pipe in accordance with the manufacturer's instructions.
- (b) Start the fan to induce an airflow down the pipe while recording the developed static air pressure and flow rate.
- (c) Slowly increase the speed of the fan until the valve opens.
- (d) Record the value of the pressure at the instant of opening—this is the opening pressure.
- (e) Adjust the airflow rate so that the measured pressure is  $-250 \pm 10$  Pa.
- (f) Measure and record the airflow rate—this is the 'determined air flow capacity'.
- (g) Reduce the airflow rate so that the measured pressure is  $-150 \pm 10$  Pa.
- (h) Measure and record the airflow rate.

#### A5 CRITERIA

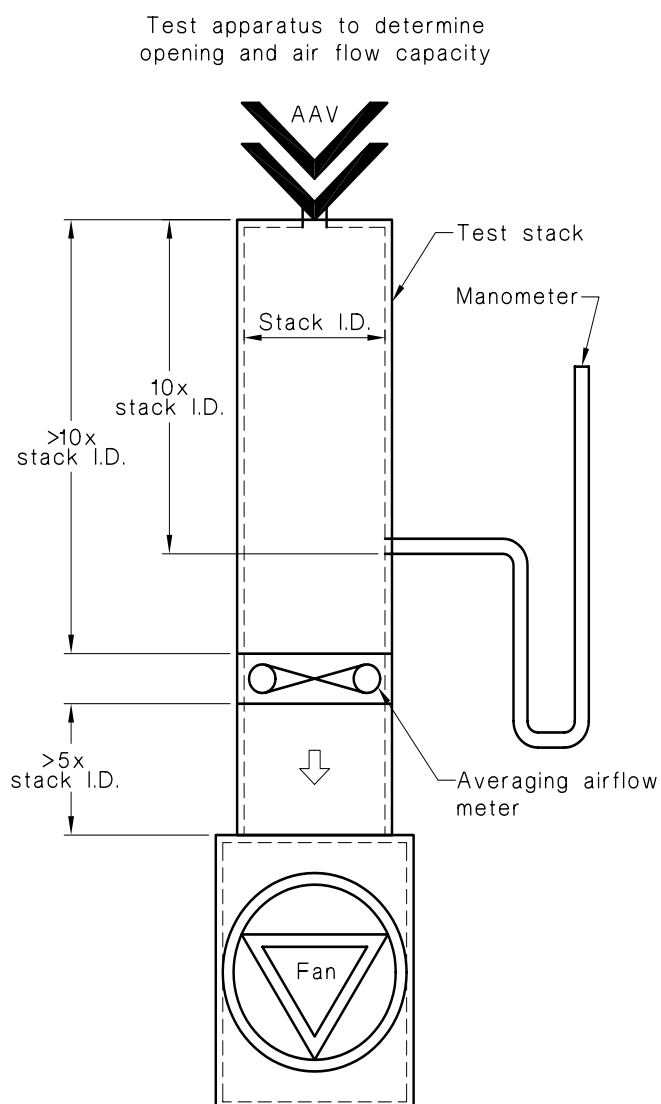
The opening pressure shall be between 0 and  $-150$  Pa. There shall be a measurable airflow rate when the applied pressure is  $-150$  Pa.

#### A6 TEST REPORT

The following shall be reported:

- (a) Manufacturer, model, type and size of valve.

- (b) The opening pressure.
- (c) The determined airflow rate capacity.
- (d) Whether both criteria have been met.
- (e) Reference to this test method, i.e. AS/NZS 4936, Appendix A.



## APPENDIX B

### DROP TEST

(Normative)

#### B1 SCOPE

This Appendix sets out the method for determining the resistance to deformation or breakage of air admittance valves, which might occur in handling before installation.

#### B2 PRINCIPLE

The valve is dropped from a predetermined height, at different orientations.

#### B3 PROCEDURE

This test shall be carried out over a smooth clean concrete surface (see method in Figure B1.)

Valves with removable insulation shall be tested with the insulation removed.

The procedure shall be as follows:

- (a) Ensure that the test surface is clean.
- (b) Hold the air admittance valve with its lowest point 1 +0.02, –0 m above the concrete as shown in orientation 1.
- (c) Release the air admittance valve.
- (d) Pick up the air admittance valve, change to orientation 2 and drop the air admittance valve.
- (e) Pick up the air admittance valve, change to orientation 3 and drop the air admittance valve.

#### B4 TEST REPORT

The following shall be reported:

- (a) Manufacturer, model, type and size of valve
- (b) Any cracks, breakages, deformation or other failure.
- (c) Reference to this test method, ie. AS/NZS 4936, Appendix B.

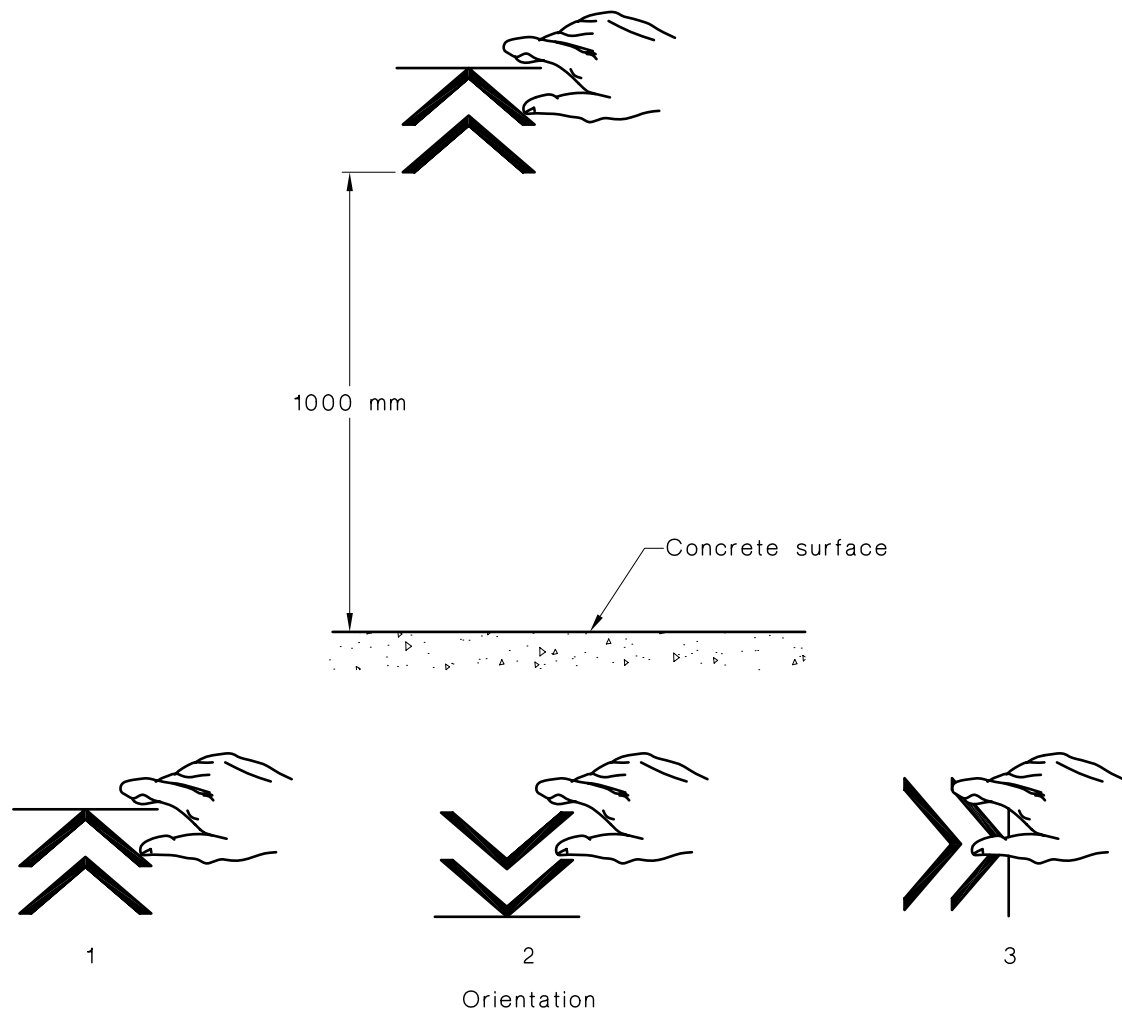


FIGURE B1 DROP TEST

## APPENDIX C

### AIRTIGHTNESS TEST

(Normative)

#### C1 SCOPE

This Appendix sets out the method for determining airtightness of air admittance valves.

#### C2 PRINCIPLE

The valve is subjected to various pressure tests, to determine leakage.

#### C3 APPARATUS

- (a) An insulated pipe of appropriate diameter and material with a length at least 2.5 times the diameter and sealed at the base (see Figure C1).
- (b) Air-isolating valve.
- (c) Air pump.
- (d) Manometer.
- (e) Means for attaching a plug and the air admittance valve to be tested so that it can be installed in accordance with the manufacturer's instructions.

##### NOTES:

- 1 Air admittance valves intended for installation by solvent welding may be solvent welded into the length of pipe and tested by fitting an airtight cap or plug to the free end. This cap or plug may carry the connections for the manometer and air inlet valve. The tests in Appendices C and D may be carried out by fitting the air admittance valve and the attached pipe to the respective test rig.
- 2 Air admittance valves integral with a fixture trap may require a suitable bend or adaptor for fitting to the test rig.
- 3 Tests should be conducted in a draught-free constant-temperature environment.

#### C4 PROCEDURE

The procedure shall be as follows:

- (a) With a plug fitted, check that the test apparatus is airtight when subject to a pressure of 10 000 +0,−100 Pa for a period of at least 5 min.
- (b) Remove the plug and fit the air admittance valve as required by the manufacturer's instructions.
- (c) Open the air inlet valve and slowly raise the air pressure to 30 +10,−0 Pa. Close the air inlet valve.
- (d) After 5 min, record the internal pressure.
- (e) If the internal air pressure is less than 90% of the initial value, check that the apparatus is still airtight. If a leak is detected in the apparatus, rectify and repeat from Step (c).
- (f) Open the air inlet valve and slowly raise the pressure to 500 +10,−0 Pa. Close the air inlet valve.
- (g) After 5 min, record the internal pressure.

- (h) If the pressure is less than 450 Pa, check that the apparatus is airtight. If a leak is detected in the apparatus, rectify and repeat from Step (c).
- (i) If the previous test criteria are achieved, raise the pressure by opening the air inlet valve and pumping air into the apparatus until it reaches 10 000 +0,–100 Pa. Close the air valve.
- (j) After 5 min, record the internal pressure.
- (k) If the test pressure is less than 9 000 Pa, check the apparatus is airtight using a soap solution and inspecting for air bubbles. If a leak is detected in the apparatus, rectify and repeat from Step (f).

## C5 CRITERIA

The pressures recorded in Clause B4 shall not be less than—

- (a) 90% of the initial value (30 +10,–0 Pa)
- (b) 450 Pa; and
- (c) 9 000 Pa, respectively.

## C6 TEST REPORT

The following shall be reported:

- (a) Manufacturer, model, type and size of valve.
- (b) Whether all criteria have been met.
- (c) Reference to this test method, i.e., AS/NZS 4936, Appendix C.

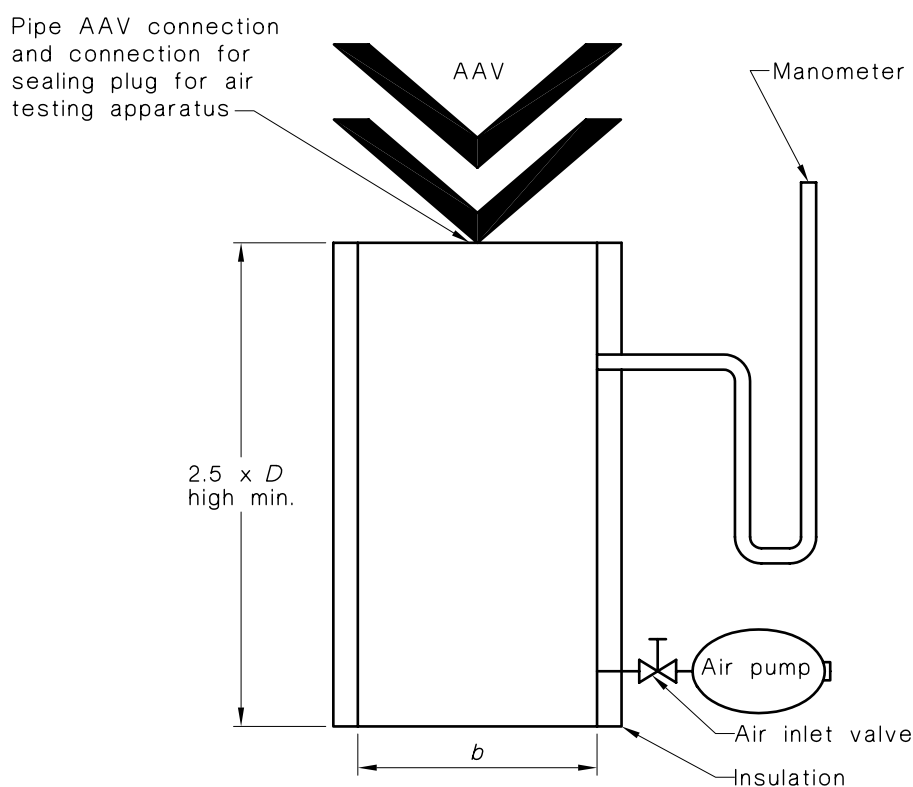


FIGURE C1 TYPICAL AIRTIGHTNESS RIG



APPENDIX D  
ENDURANCE AND TEMPERATURE TEST  
(Normative)

**D1 SCOPE**

This Appendix sets the method for determining the endurance of air admittance valves.

**D2 PRINCIPLE**

The valve is operated at various temperatures for specified periods of time.

**D3 APPARATUS**

A rig assembled according to Figure D1, comprised of—

- (a) A pipe fitted with an electric fan at one end;
- (b) A means for diverting air flow in the pipe;
- (c) Temperature measuring and pressure recording devices.

An atmospheric vent may be required to ensure that the air admittance valve closes and that the applied suction is within the specified limits.

**D4 PROCEDURE**

When supplied with an insulating cap, the cap shall be in place on the air admittance valve. Tests are conducted on only one valve randomly selected from the three sample valves.

The procedure shall be as follows:

- (a) The air admittance valve and test apparatus shall be placed and operated in an environment where the ambient air is maintained at  $20 \pm 5^\circ\text{C}$  for a period of  $480 \pm 5$  min.
- (b) Upon completion of this test, conduct again the airtightness test given in Appendix C.
- (c) The air admittance valve and test apparatus shall be placed and operated in an environment where the ambient air is maintained at  $60 +0, -10^\circ\text{C}$  for a period of  $480 \pm 5$  min.
- (d) Upon completion of this test, conduct again the airtightness test given in Appendix C.

**D5 CRITERIA**

The means of diverting airflow shall open and close the air admittance valve  $15 \pm 2$  times each minute and result in the specified pressure/time functions shown in Figure D2.

**D6 TEST REPORT**

The following shall be reported:

- (a) Manufacturer, model, type and size of valve.
- (b) Whether all criteria have been met.
- (c) Reference to this test method, ie AS/NZS 4936, Appendix D.

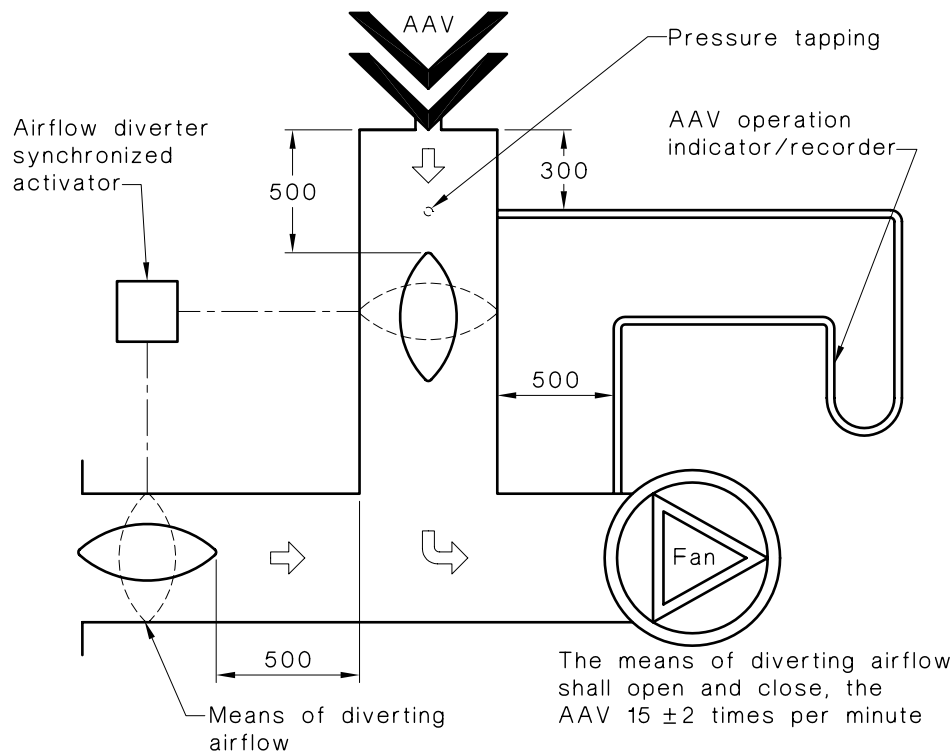


FIGURE D1 TYPICAL ENDURANCE AND TEMPERATURES RIG

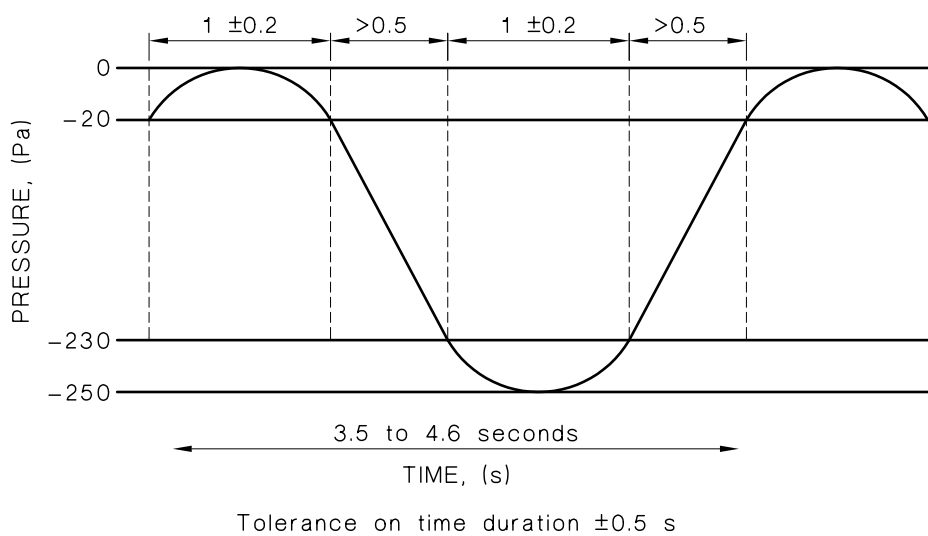


FIGURE D2 TIME/PRESSURE GRAPH

NOTE: An atmosphere vent may be required to ensure that the air admittance valve closes and that the applied suction is within the specified limits.

## APPENDIX E

### INSTALLATION INSTRUCTIONS

(Normative)

Each valve shall be packaged with a complete set of installation instructions that shall provide all information necessary for the correct installation of the valve.

Where applicable, the instructions shall include the following information:

- (a) The maximum capacity of the valve, expressed in fixture units, being the values from Table 1 corresponding to the determined airflow capacity, and which state the maximum loading of—
  - (i) the discharge pipe to which it may be connected when used for trap or group venting; and
  - (ii) the discharge stack when used for stack venting.
- (b) Permissible locations for the valve.
- (c) Orientation of the valve.
- (d) Minimum clearances from the pipe obvert.
- (e) Jointing methods or method of connection to the vent pipe.
- (f) Necessary clearances to allow access for inspection, maintenance, testing and replacement.

**TABLE 7.1**  
**CAPACITY OF VALVES**

| Determined airflow capacity,<br>L/s | Maximum fixture unit loading<br>for discharge pipes | Maximum fixture unit loading<br>for discharge stacks |
|-------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| 1                                   | 6                                                   | —                                                    |
| 2                                   | 8                                                   | —                                                    |
| 2.5                                 | 10                                                  | —                                                    |
| 3                                   | 15                                                  | —                                                    |
| 4                                   | 27                                                  | 1                                                    |
| 6                                   | 60                                                  | 3                                                    |
| 8                                   | 108                                                 | 6                                                    |
| 10                                  | 168                                                 | 10                                                   |
| 12                                  | 243                                                 | 15                                                   |
| 15                                  | 379                                                 | 23                                                   |
| 20                                  | —                                                   | 42                                                   |
| 25                                  | —                                                   | 65                                                   |
| 30                                  | —                                                   | 94                                                   |
| 40                                  | —                                                   | 168                                                  |
| 60                                  | —                                                   | 379                                                  |
| 80                                  | —                                                   | 675                                                  |
| 100                                 | —                                                   | 1054                                                 |

NOTE: Intermediate values may be calculated from the following equations on which this table is based and rounded down to the nearest integer value.

For discharge pipes with fixture unit values greater than 8 the maximum fixture unit loading at the point of connection is equal to:

$$6.75 (Q/2)^2,$$

For discharge stacks the maximum fixture unit loading at base of stack is equal to:

$$6.75 (Q/8)^2$$

Q is the determined air flow capacity in litres per second

NOTES

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